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FIELD NOTE · BATTERY SETTINGS

Harmony @Huailan: battery settings changed to buy less from the grid

Solar ESS Working Group · 2 Nov 2026 · 10 kW hybrid solar + battery system, Chiang Mai

Our review of this eco-lodge's electricity use showed the battery settings were costing money. On 2 November 2026 we changed them remotely through the monitoring platform. Here is what changed, in plain words.

What we found

- **Battery was held at 40%** most of the day and night, and pushed up to **80% from 09:00 to 17:00**.
- **Grid charging was on in every time slot**, so the inverter bought grid power in the morning to fill the battery, at the very time the solar panels were about to do it for free.
- Result: **more grid electricity bought** than needed, and only part of the battery was ever used.

What we changed

- **Battery can now run down to 25%** in every time slot before the house draws from the grid.
- **Grid charging is off** in every slot: the battery is now charged by the sun only.
- Time of Use stays on; slot times are unchanged. Selling to the grid stays off.

Time slot	Battery level before	Battery level now	Grid charge before	Grid charge now
01:00 – 05:00	40%	25%	On	Off
05:00 – 09:00	40%	25%	On	Off
09:00 – 13:00	80%	25%	On	Off
13:00 – 17:00	80%	25%	On	Off
17:00 – 21:00	40%	25%	On	Off
21:00 – 01:00	40%	25%	On	Off

The inverter confirmed the new settings at 09:00 on 2 November 2026 (command state: succeeded).

What the owner should see

- **Less grid electricity bought**, especially in the early morning.
- **More of the battery is used** every night (from 25% to full instead of 40–80%).
- Trade-off: during a night-time blackout, there may be as little as 25% left in the battery. The owner can raise the night slots to 40% if backup matters more than savings.

Next step: Solar ESS will compare the next days of monitoring data with the weeks before the change to confirm the saving.

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